

ROTTERDAM THE HAGUE AP, Netherlands

WMO: 063440

Lat: 51.961N

Lon: 4.447E

Elev: -15

StdP: 14.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.3	24.9	14.5	11.5	23.9	18.9	14.2	26.7	32.6	48.5	29.8	49.2	6.9	60	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	83.1	68.1	79.0	66.6	75.5	64.8	69.8	79.8	67.9	76.4	66.2	73.5	8.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.3	73.8	64.8	91.7	71.6	63.3	86.9	69.8	33.8	79.9	32.2	76.7	30.9	73.7	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	23.2	20.5	DB	16.3	90.2	5.4	3.1	12.5	92.4	9.3	94.3	6.3	96.0	2.4	98.3	(4)
			WB	15.5	73.6	5.2	1.6	11.8	74.7	8.7	75.6	5.8	76.5	1.9	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	39.0	39.7	43.7	49.4	55.5	60.5	64.7	64.3	59.2	52.9	45.6	40.2	(6)
		DBStd	11.00	7.20	6.77	5.82	6.33	6.19	5.36	5.21	4.66	4.65	5.78	5.98	7.28	(7)
		HDD50	1444	343	290	206	88	17	0	0	0	37	154	309		(8)
		HDD65	5150	807	708	661	470	303	154	69	70	182	376	581	769	(9)
		CDD50	1913	1	2	10	68	187	316	456	443	277	126	22	5	(10)
		CDD65	145	0	0	0	0	7	20	60	48	9	1	0	0	(11)
		CDH74	1085	0	0	0	7	75	185	448	319	49	2	0	0	(12)
		CDH80	262	0	0	0	0	10	42	125	78	7	0	0	0	(13)
(14)	Wind	WSAvg	9.6	11.7	11.2	10.8	9.1	8.9	8.6	8.2	7.9	8.2	9.5	9.8	11.0	(14)
(15)	Precipitation	PrecAvg	35.5	3.0	2.7	2.0	1.7	2.2	2.8	3.4	3.9	3.3	3.3	3.7	3.4	(15)
(16)		PrecMax	47.6	6.7	5.2	4.2	4.2	4.7	5.9	7.4	9.5	9.6	6.7	6.3	6.6	(16)
(17)		PrecMin	25.1	0.3	0.7	0.4	0.0	0.9	1.0	0.6	0.8	0.7	0.6	0.3	0.6	(17)
(18)		PrecStd	5.2	1.5	1.1	1.1	1.0	1.1	1.4	1.7	2.2	2.1	1.5	1.5	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.5	56.9	63.9	74.6	80.9	85.3	89.1	86.8	79.4	71.5	61.3	55.9		(19)
(20)		0.4% MCWB	51.8	49.9	54.4	59.6	65.7	68.6	69.5	69.8	67.1	62.9	57.3	53.5		(20)
(21)		2% DB	52.0	52.9	58.9	68.9	75.5	79.3	83.4	81.8	73.9	66.2	58.2	54.1		(21)
(22)		2% MCWB	49.8	49.0	51.6	57.1	61.7	66.3	68.3	67.9	65.0	60.8	55.4	52.0		(22)
(23)		5% DB	50.1	50.8	55.0	64.2	71.3	74.8	78.7	77.3	70.1	63.2	56.2	52.1		(23)
(24)		5% MCWB	47.9	47.9	49.9	54.6	59.9	64.2	66.8	66.4	62.9	59.1	53.5	50.1		(24)
(25)		10% DB	48.2	48.9	52.4	59.9	67.1	70.8	74.9	73.3	67.1	61.1	54.0	49.9		(25)
(26)		10% MCWB	46.1	46.4	48.4	52.8	58.3	62.0	65.1	64.7	61.2	57.5	51.5	47.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	52.2	51.7	55.4	61.8	67.3	70.9	72.3	72.0	68.6	63.9	58.1	54.2		(27)
(28)		0.4% MCDB	53.7	54.7	61.4	72.2	78.2	81.5	82.8	82.5	76.7	69.5	60.1	55.4		(28)
(29)		2% WB	50.2	49.7	52.7	57.9	63.7	67.5	69.8	69.6	66.1	61.4	55.9	52.2		(29)
(30)		2% MCDB	51.6	51.8	57.2	66.5	72.0	76.4	80.1	78.8	72.3	65.3	58.1	53.8		(30)
(31)		5% WB	48.1	48.2	50.6	55.6	61.0	65.1	67.9	67.5	64.0	59.9	53.9	50.4		(31)
(32)		5% MCDB	49.7	50.4	54.3	62.8	69.0	72.9	76.7	74.9	68.7	63.0	55.9	51.9		(32)
(33)		10% WB	46.4	46.6	48.9	53.4	58.9	62.9	66.0	65.7	62.3	58.1	51.9	48.1		(33)
(34)		10% MCDB	48.1	48.8	52.0	59.0	66.2	69.2	73.3	71.8	66.2	60.7	53.7	49.8		(34)
(35)	Mean Daily Temperature Range	MDBR	8.1	9.6	12.3	15.2	15.6	15.3	14.8	14.8	13.5	11.7	9.2	8.1		(35)
(36)		5% DB MCDBR	8.2	10.7	17.1	21.8	23.0	22.9	22.1	21.6	17.6	13.2	9.4	7.4		(36)
(37)		5% WB MCWBR	7.9	8.3	11.4	12.4	11.9	11.9	10.2	10.3	10.0	9.0	8.0	7.1		(37)
(38)		5% DB MCDBR	7.9	9.4	14.7	19.5	20.1	19.6	19.4	18.8	15.6	12.6	9.1	7.4		(38)
(39)		5% WB MCWBR	7.9	8.0	10.6	11.8	11.5	11.2	10.0	10.0	9.5	9.2	8.2	7.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.363	0.397	0.423	0.430	0.425	0.433	0.424	0.401	0.377	0.354	0.338		(40)
(41)		taud	2.316	2.302	2.222	2.190	2.228	2.262	2.248	2.289	2.342	2.364	2.334	2.306		(41)
(42)		Ebn at Noon	204	235	249	257	261	264	259	255	246	227	197	183		(42)
(43)		Edh at Noon	21	28	36	42	42	41	41	37	32	26	20	18		(43)
(44)	All-Sky Solar Radiation	RadAvg	226	444	826	1334	1615	1730	1635	1379	1003	578	264	171		(44)
(45)		RadStd	25	83	104	134	115	123	180	115	87	48	25	20		(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page